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Supplementary Material

Higher methanotroph abundance and bottom-water methane in ponds with floating photovoltaic arrays

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Supplemental Methods

Water Column Microbial Cell Abundances

Water samples were preserved for flow cytometry by adding 1 µL of 25% glutaraldehyde to 1 mL of 20 µm pre-filtered pond water, shaking, incubating for 10 minutes in the dark at room temperature, and freezing at -80 °C. Fixed samples were diluted to 15-20X, stained with SYBR Green I, and run in triplicate on an Attune NxT Flow Cytometer. DNA-positive cell counts were analyzed using the *flowCore* (Ellis et al. 2025) and *ggcyto* (Van et al. 2018) packages in R (R Core Team 2014).

Testing for Water Column Stratification

We tested if the water column was stratified for each pond on each sampling date by comparing calculated density gradients between ponds with and without FPV. We used temperature measurements in the surface and bottom waters, corrected for the distance between those samples. Water density at a given temperature was calculated as:

$$\rho(T) = \frac{\rho_{H_2O}}{1 + \beta(T - T_0)} \times \frac{1000 \text{ kg m}^{-3}}{1 \text{ kg L}^{-1}}$$

Where $\rho(T)$ is the density of water at a given temperature (kg m^{-3}), ρ_{H_2O} is the nominal density of water (assuming 1 kg L^{-1} for freshwater), β is the volumetric temperature expansion coefficient ($0.0002 \text{ }^\circ\text{C}^{-1}$), T is the temperature of the water (C), and T_0 is standard temperature ($20 \text{ }^\circ\text{C}$).

We then calculated the density gradient between the surface and bottom water temperature ($\text{kg m}^{-3} \text{ m}^{-1}$) for each pond on each sampling day, and compared density gradients between ponds with and without FPV by constructing a mixed effect model with FPV presence and day of year as fixed effects and pond as a random effect, using least square means to compare density gradients between ponds with and without FPV.

References

- Ellis, B., P. Haaland, Hahne F, N. Le Meur, N. Gopalakrishnan, J. Spidlen, M. Jiang, and G. Finak. 2025. *flowCore*: *flowCore*: Basic structures for flow cytometry data. R package version 2.2.0.
- Van, P., W. Jiang, R. Gottardo, and G. Finak. 2018. *ggcyto*: Next-generation open-source visualization software for cytometry. *Bioinformatics* **34**: 3951–3953.
- R Core Team. 2014. R: A language and environment for statistical computing.

Table S1: Methanogen and Methanotroph Taxonomic Classification

Order	Family	Genus	Methane Cycler
<i>Methanocellales</i>	<i>Methanocellaceae</i>	<i>Methanocella_A</i>	Methanogen
<i>Methanomicrobiales</i>	<i>Methanomicrobiaceae</i>		Methanogen
<i>Methanomicrobiales</i>	<i>Methanospirillaceae_2121</i>	<i>Methanolinea_A</i> <i>Methanolinea_B</i> <i>Methanoregula</i> <i>UBA467</i> <i>UBA288</i>	Methanogen
<i>Methanomicrobiales</i>	<i>Methanospirillaceae_2125</i>	<i>Methanospirillum</i>	Methanogen
<i>Methanosarcinales_A_2632</i>	<i>Methanosarcinaceae</i>	<i>Methanosarcina_2619</i> <i>Methanomethylovorans</i>	Methanogen
<i>Methanotrichales</i>	<i>Methanotrichaceae</i>	<i>Methanothrix_B</i>	Methanogen
<i>Methanobacteriales</i>	<i>Methanobacteriaceae</i>	<i>Methanobacterium_A</i> <i>Methanobacterium_B_963</i> <i>Methanobacterium_C</i> <i>Methanobacterium_D_1054</i> <i>Methanobacterium_F_896</i> <i>Methanobacterium_F_900</i> <i>Methanobrevibacter_A</i> <i>Methanobrevibacter_D</i> <i>Methanosphaera</i>	Methanogen
<i>Methanofastidiosales</i>	<i>Methanofastidiosaceae</i>	<i>Methanofastidiosum</i>	Methanogen
<i>Methanomassiliicoccales</i>	<i>Methanomassiliicoccaceae</i>	<i>Methanomassiliicoccus_A_1624</i>	Methanogen
<i>Methanomassiliicoccales</i>	<i>Methanomethylophilaceae</i>		Methanogen

<i>Methanomassiliicoccales</i>	<i>UBA472</i>	<i>FEN-33</i>	Methanogen
<i>Methanomethylicaes</i>	<i>Methanomethylicaceae</i>	<i>Methanomethylicus</i>	Methanogen
<i>Methanosarcinales_A_2632</i>	<i>Methanoperedenaceae</i>	<i>Methanoperedens_A</i>	Methanotroph
<i>Methyloirabilales</i>	<i>2-02-FULL-66-22</i>	<i>2-02-FULL-66-22</i>	Methanotroph
<i>Rhizobiales_505101</i>	<i>Beijerinckiaceae</i>	<i>Methylocystis</i> <i>Methylosinus</i>	Methanotroph
<i>Methylococcales</i>	<i>Methylococcaceae</i>	<i>Methylocaldum</i> <i>Methylococcus</i> <i>Methylomagnum</i> <i>Methyloparacoccus</i> <i>Methyloterricola</i> <i>Methylotetracoccus</i> <i>UBA6136</i>	Methanotroph
<i>Methylococcales</i>	<i>Methylomonadaceae</i>	<i>Crenothrix</i> <i>Methylobacter_C_601048</i> <i>Methylobacter_C_601751</i> <i>Methyloglobulus</i> <i>Methylomonas</i> <i>Methylosoma</i> <i>Methylovulum</i> <i>UBA4132</i>	Methanotroph

Table S2: Summary of model information for mixed models to describe temperature, dissolved oxygen, surface and bottom water dissolved methane (CH₄) concentration, methanotroph abundance, and methanogen abundance, relative abundance of sediment methanogens and methanotrophs, and density gradients in ponds with and without floating photovoltaics (FPV). All models include FPV presence or absence and day of year as fixed effects and the pond ID as a random effect. For each fixed effect, the estimated effect size is listed $\pm$ standard error of the effect, with the associated t-value below in parentheses.

Response	Observations (n)	Intercept	FPV Presence	Day of Year
Temperature	36	24.9 $\pm$ 2.06 (12.1)	3.91 $\pm$ 0.60 (6.46)	-0.03 $\pm$ 0.01 (-3.00)
Dissolved Oxygen (% Saturation)	36	-19.2 $\pm$ 18.5 (-1.03)	32.2 $\pm$ 5.43 (5.92)	0.23 $\pm$ 0.08 (2.72)
Dissolved Oxygen (mg L ⁻¹)	36	-3.45 $\pm$ 1.61 (-2.14)	2.72 $\pm$ 0.53 (5.13)	0.03 $\pm$ 0.01 (3.93)
Surface Water [CH ₄]	36	5.61 $\pm$ 2.60 (2.16)	-0.82 $\pm$ 0.65 (-1.26)	-0.01 $\pm$ 0.01 (-0.62)
Bottom Water [CH ₄]	36	631 $\pm$ 289 (2.18)	-373 $\pm$ 126 (-2.96)	-1.02 $\pm$ 1.33 (-0.77)
Surface water methanotroph abundance	24	-626754 $\pm$ 287292 (-2.18)	-318325 $\pm$ 86085 (-3.70)	4997 $\pm$ 1316 (3.80)
Surface water methanogen abundance	24	-1332 $\pm$ 2632 (0.51)	-125 $\pm$ 789 (-0.16)	0.44 $\pm$ 12.05 (0.04)
Bottom water methanotroph abundance	24	-94652 $\pm$ 324453 (-0.29)	-291441 $\pm$ 97220 (-3.00)	2623 $\pm$ 1485 (1.77)
Bottom water methanogen abundance	24	24108 $\pm$ 28147 (0.40)	8895 $\pm$ 12441 (0.51)	-63.13 $\pm$ 125 (-0.62)
Sediment methanotroph abundance	23	0.15 $\pm$ 0.03 (5.51)	0.007 $\pm$ 0.008 (0.86)	-0.000 $\pm$ 0.000 (-3.42)
Sediment methanogen abundance	23	0.32 $\pm$ 0.05 (5.83)	0.014 $\pm$ 0.030 (0.46)	-0.001 $\pm$ 0.000 (-2.47)
Density Gradient (kg m ⁻³ m ⁻¹)	36	3.53 $\pm$ 0.37 (9.54)	-0.145 $\pm$ 0.108 (-1.34)	-0.013 $\pm$ (0.002) (-7.65)

Table S3: Summary of model information for mixed models to describe bottle incubations to determine rates of water column dissolved methane (CH₄) production or consumption in ponds with and without floating photovoltaics (FPV). All models include FPV presence or absence and if the bottle was light or dark as fixed effects and the pond ID as a random effect. For each fixed effect, the estimated effect size is listed $\pm$ standard error of the effect, with the associated t-value below in parentheses.

Response	Observations (n)	Intercept	FPV Presence/Absence	Light/Dark Bottle
Surface CH ₄ Flux	12	-0.09 $\pm$ 0.07 (-1.27)	-0.08 $\pm$ 0.10 (-0.82)	0.08 $\pm$ 0.03 (3.01)
Bottom CH ₄ Flux	12	-1.69 $\pm$ 4.14 (-0.41)	2.11 $\pm$ 5.83 (0.36)	-0.66 $\pm$ 0.88 (-0.75)

Table S4: Results of least-square means comparisons between ponds with and without floating photovoltaics. Models for which comparisons are made are described in Tables S1 and S2. For bottle incubations to assess water column methane transformations, a positive value indicates net methane production, while a negative value indicates net methane consumption.

Comparison	Estimate $\pm$ SE	z-ratio	p-value
Temperature ($^{\circ}\text{C}$)	-3.91 ± 0.60	-6.46	< 0.001
Dissolved Oxygen (% Saturation)	-32.3 ± 5.43	-5.92	< 0.001
Dissolved Oxygen (mg L^{-1})	-2.72 ± 0.53	-5.13	< 0.001
Surface Water [CH_4]	0.82 ± 0.65	1.26	0.208
Bottom Water [CH_4]	373 ± 126	2.96	0.003
Surface Water CH_4 Production/Consumption	0.08 ± 0.10	0.82	0.413
Bottom CH_4 Production/Consumption	-2.11 ± 5.83	-0.36	0.718
Surface water methanotroph abundance	318325 ± 86100	3.70	0.021
Surface water methanogen abundance	125 ± 789	0.158	0.882
Bottom water methanotroph abundance	291441 ± 97200	3.00	0.040
Bottom water methanogen abundance	-8895 ± 12400	-0.72	0.514
Sediment methanotroph abundance	-0.007 ± 0.008	-0.858	0.440
Sediment methanogen abundance	-0.014 ± 0.030	-0.46	0.667
Density Gradient	0.145 ± 0.108	1.40	0.180

Table S5: Results of Welch two-sample t-tests comparing diffusive methane fluxes and gas exchange velocities (k600) between the center and edges of ponds with and without floating photovoltaic arrays (FPV). We first compared pond locations across treatments, and then between locations and within treatments.

Comparisons Between Ponds with and Without FPV				
Measurement	Pond Location	df	t-value	p-value
Diffusive CH ₄ Flux	Center	2.00	-1.03	0.413
Diffusive CH ₄ Flux	Edge	2.00	-0.97	0.436
k600	Center	2.18	-0.75	0.525
k600	Edge	2.00	-1.03	0.412
Comparisons between pond Center and Edge within the same pond type				
Measurement	FPV Present/Absent	df	t-value	p-value
Diffusive CH ₄ Flux	Present	2.28	-1.67	0.221
Diffusive CH ₄ Flux	Absent	3.34	0.34	0.751
k600	Present	2.03	1.17	0.362
k600	Absent	3.42	-0.31	0.774

Table S6: Results of Welch two-sample t-test comparing potential rates of sediment CH₄ production between ponds with and without floating photovoltaic arrays.

Measurement	df	t-value	p-value
Sediment CH ₄ Flux	2.75	-1.26	0.302

Table S7: Summary of mean $\pm$ standard deviation of measured concentrations and rates in ponds with FPV and control ponds without FPV. For potential sediment CH₄ production and water column CH₄ “flux”, positive values indicate net-CH₄ production while negative values indicated net-consumption.

Measurement	Control Ponds	FPV Ponds
Temperature (°)	22.78 $\pm$ 2.29	18.87 $\pm$ 1.69
Dissolved Oxygen (% Saturation)	62.34 $\pm$ 22.79	30.18 $\pm$ 10.56
Dissolved Oxygen (mg L ⁻¹)	5.42 $\pm$ 2.19	2.70 $\pm$ 0.98
Surface Water [CH ₄] (μ mol L ⁻¹)	3.19 $\pm$ 2.07	4.01 $\pm$ 1.79
Bottom Water [CH ₄] (μ mol L ⁻¹)	45.6 $\pm$ 129.7	419 $\pm$ 306
Surface water methanotroph abundance		
Surface water methanogen abundance		
Bottom water methanotroph abundance		
Bottom water methanogen abundance		
Sediment methanotroph abundance		
Sediment methanogen abundance		
Density Gradient (kg m ⁻³ m ⁻¹)	0.62 $\pm$ 0.42	0.76 $\pm$ 0.63
Potential Sediment CH ₄ Production (ppm CH ₄ g dry sediment ⁻¹ d ⁻¹)	14.1 $\pm$ 10.9	5.43 $\pm$ 4.81
Surface Water Column CH ₄ “Flux” (μ mol CH ₄ L ⁻¹ hr ⁻¹)	-0.14 $\pm$ 0.15	-0.05 $\pm$ 0.08
Bottom Water Column CH ₄ “Flux” (μ mol CH ₄ L ⁻¹ hr ⁻¹)	0.09 $\pm$ 1.58	-2.02 $\pm$ 9.03

Table S8: FPV treatment, pond, and day of year (DOY) significantly influence the combined methanogen and methanotroph community composition in the water column and sediments. Bray-Curtis dissimilarities were calculated using absolute abundances for water (Fig. 3A) and scaled relative abundances for sediment (Fig. 3B). PERMANOVA tested effects of FPV, pond, and DOY and their interactions on community composition, with the pseudo-F and R^2 representing group separation variance explained, respectively. β -dispersion tested group variance. Residual R^2 shown for each habitat.

Habitat Type	Variable	PERMANOVA				β -Dispersion		Figure
		R^2	Pseudo-F	p-value	Residuals	F	p-value	
Water	FPV	0.128	9.42	0.001	0.489	10.23	0.002	3A
Water	Pond	0.155	2.85	0.001		1.20	0.305	3A
Water	DOY	0.086	6.35	0.001		5.26	0.005	3A
Water	FPV:DOY	0.059	4.34	0.001				3A
Water	Pond:DOY	0.083	1.52	0.041				3A
Sediment	FPV	0.135	12.02	0.001	0.361	1.51	0.216	3B
Sediment	Pond	0.285	6.32	0.001		0.66	0.68	3B
Sediment	DOY	0.107	9.48	0.001		1.73	0.172	3B
Sediment	FPV:DOY	0.030	2.67	0.018				3B
Sediment	Pond:DOY	0.082	1.81	0.011				3B

Table S9: FPV, pond, and day of year (DOY) structure sediment methanogen and methanotroph communities with distinct drivers and effect strengths. PERMANOVA and β -dispersion results for sediment methanogen and methanotroph communities. Bray-Curtis dissimilarities were calculated from rarefied relative abundances. FPV, pond, and DOY were modeled as main effects and interactions. R^2 indicates the proportion of variance explained; pseudo-F and p-values assess compositional differences. β -dispersion statistics test group variability. Residual R^2 is reported for each CH₄ group. Figures S3A and S3B show corresponding ordinations.

Habitat Type	CH ₄ Group	Variable	PERMANOVA				β -Dispersion		Figure
			R ²	Pseudo-F	p-value	Residuals	F	p-value	
Sediment	Methanogens	FPV	0.135	12.4	0.001	0.349	2.92	0.105	S3A
Sediment	Methanogens	Pond	0.305	7.01	0.001		0.809	0.557	S3A
Sediment	Methanogens	DOY	0.098	9.01	0.001		1.03	0.411	S3A
Sediment	Methanogens	FPV:DOY	0.033	3.06	0.012				S3A
Sediment	Methanogens	Pond:DOY	0.079	1.82	0.016				S3A
Sediment	Methanotrophs	FPV	0.131	10.5	0.001	0.398	0.004	0.937	S3B
Sediment	Methanotrophs	Pond	0.231	4.63	0.001		0.723	0.604	S3B
Sediment	Methanotrophs	DOY	0.133	10.7	0.001		4.13	0.009	S3B
Sediment	Methanotrophs	FPV:DOY	0.020	1.59	NS				S3B
Sediment	Methanotrophs	Pond:DOY	0.088	1.76	0.018				S3B

Table S10: Summary of diffusive CH₄ fluxes, dissolved CH₄ concentrations, and calculated k₆₀₀ values for the edges and centers of ponds with and without FPV. Values reported as mean ± standard deviation.

Measurement	Location	Control	FPV
Diffusive CH ₄ Flux ($\mu\text{mol m}^{-2} \text{ hr}^{-1}$)	Center	954 ± 1565	26.7 ± 5.6
Diffusive CH ₄ Flux ($\mu\text{mol m}^{-2} \text{ hr}^{-1}$)	Edge	588 ± 969	47.8 ± 21.1
Dissolved CH ₄ Concentration ($\mu\text{mol L}^{-1}$)	Center	5.99 ± 3.43	2.02 ± 1.44
Dissolved CH ₄ Concentration ($\mu\text{mol L}^{-1}$)	Edge	3.08 ± 1.03	6.18 ± 2.47
k ₆₀₀ (cm h ⁻¹)	Center	0.40 ± 0.62	0.12 ± 0.13
k ₆₀₀ (cm h ⁻¹)	Edge	0.60 ± 0.96	0.03 ± 0.01

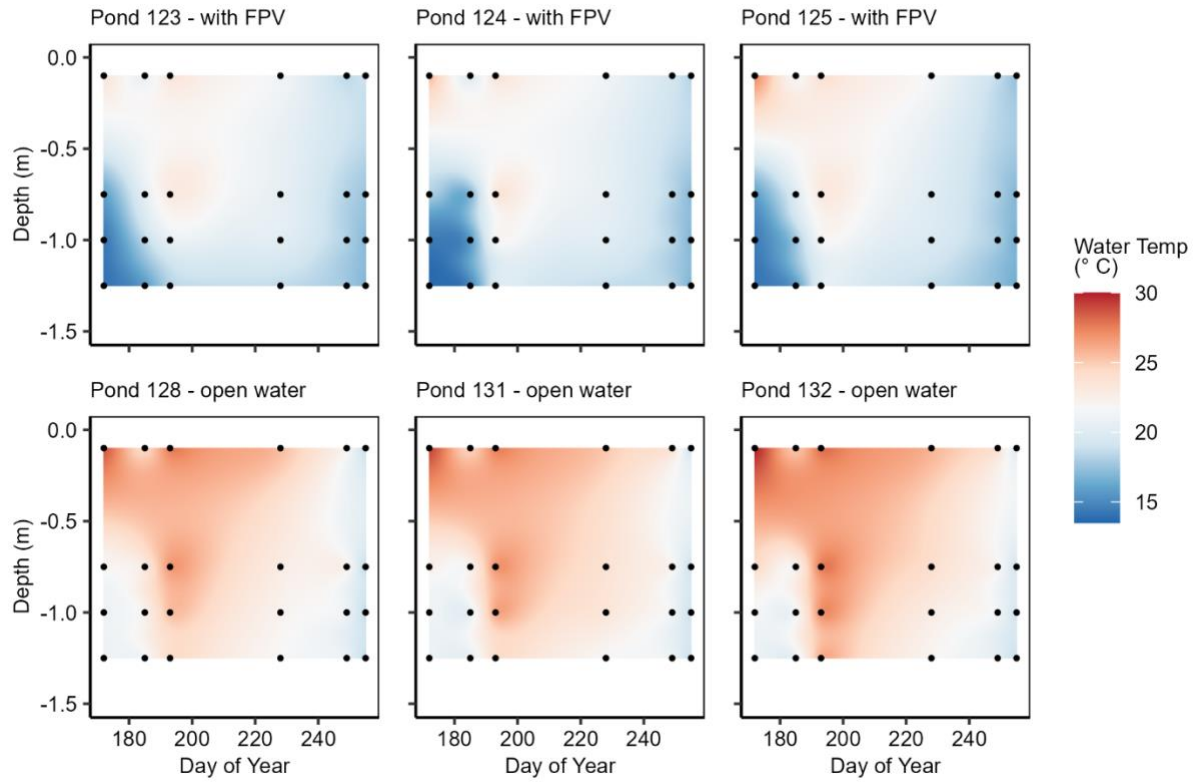


Fig. S1: Temperature profiles throughout summer 2024 in ponds with floating photovoltaic (FPV) arrays (top row) and ponds without FPV arrays (bottom row). Points indicate measurements and colors are interpolated based on those measurements.

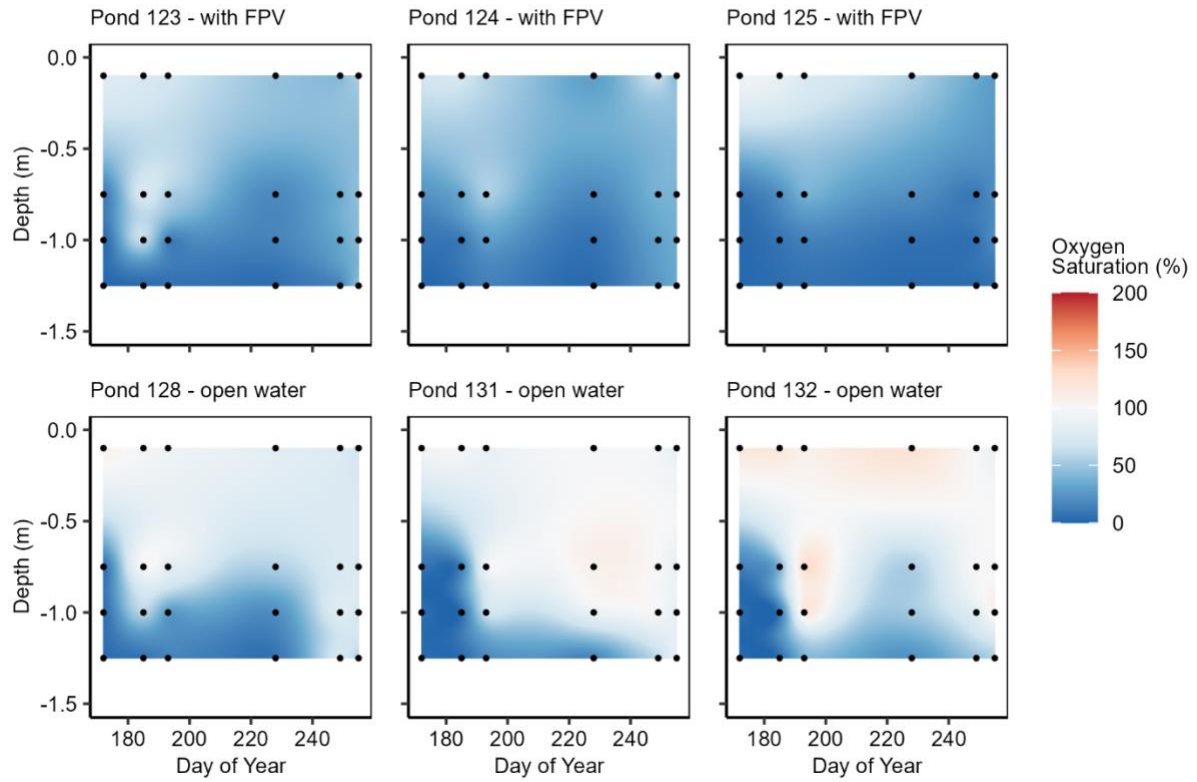


Fig. S2: Dissolved oxygen (percent saturation) profiles throughout summer 2024 in ponds with floating photovoltaic (FPV) arrays (top row) and ponds without FPV arrays (bottom row). Points indicate measurements and colors are interpolated based on those measurements.

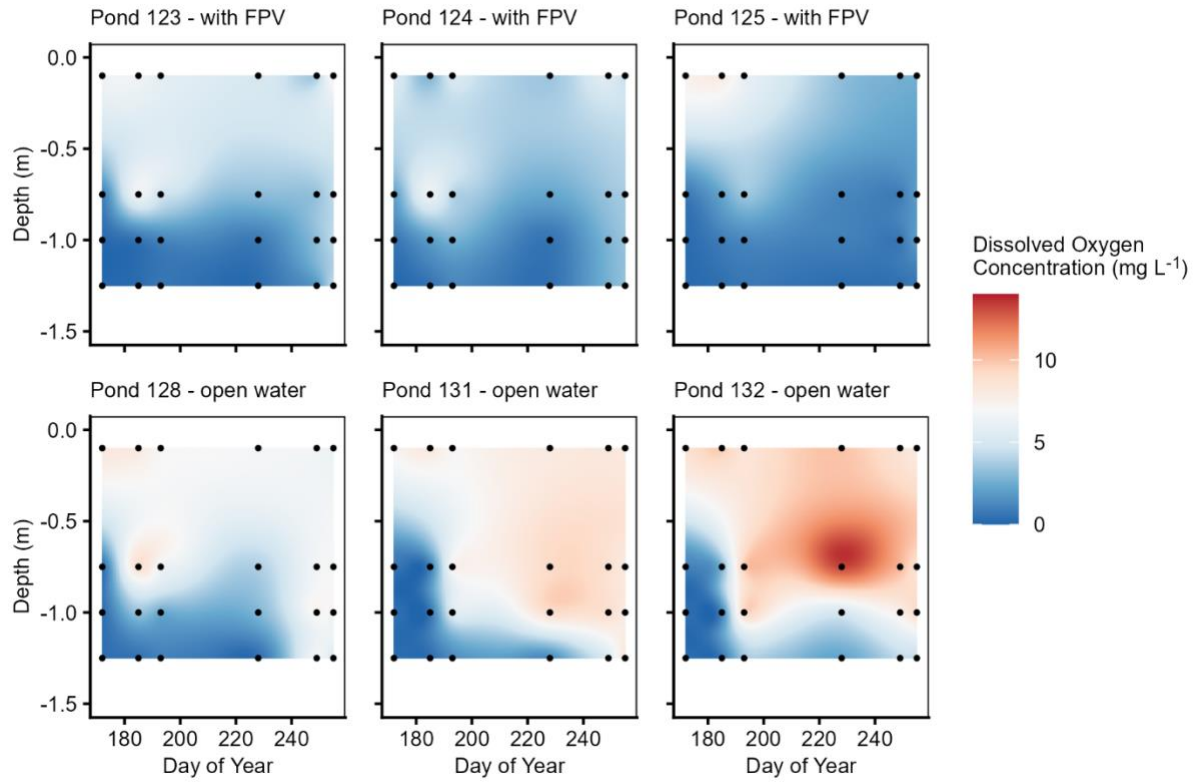


Fig. S3: Dissolved oxygen (mg L⁻¹) profiles throughout summer 2024 in ponds with floating photovoltaic (FPV) arrays (top row) and ponds without FPV arrays (bottom row). Points indicate measurements and colors are interpolated based on those measurements.

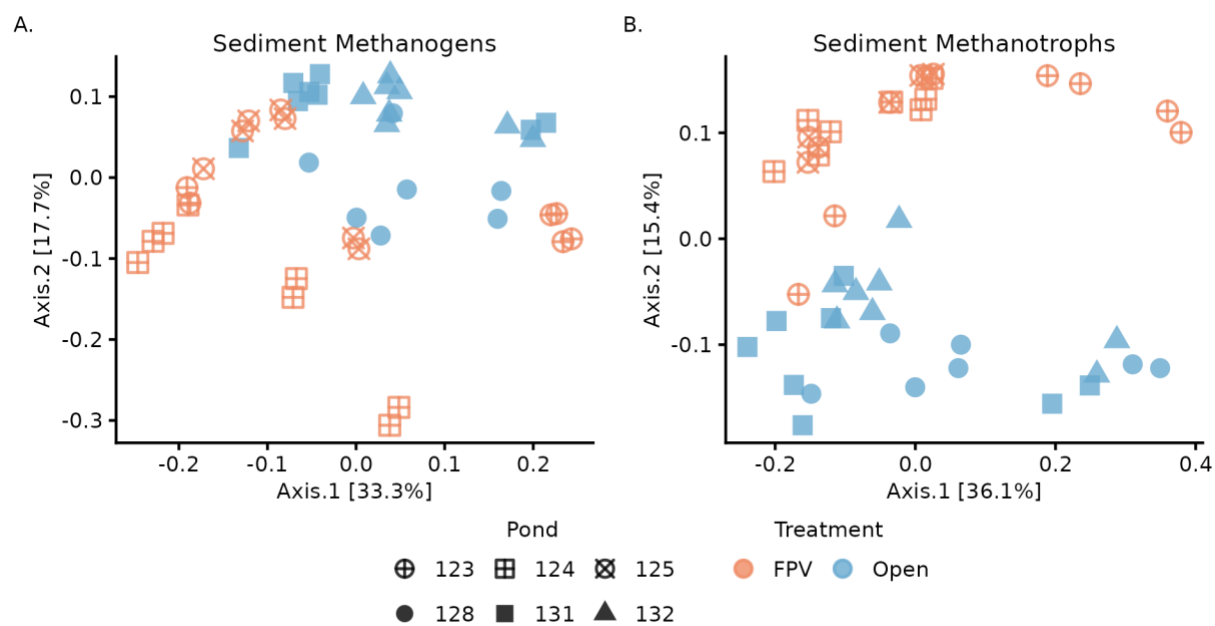


Fig. S4: Bray-Curtis PCoA of sediment (A) methanogenic and (B) methanotrophic communities within FPV and Open ponds; Sediment replicates and are shown separately.

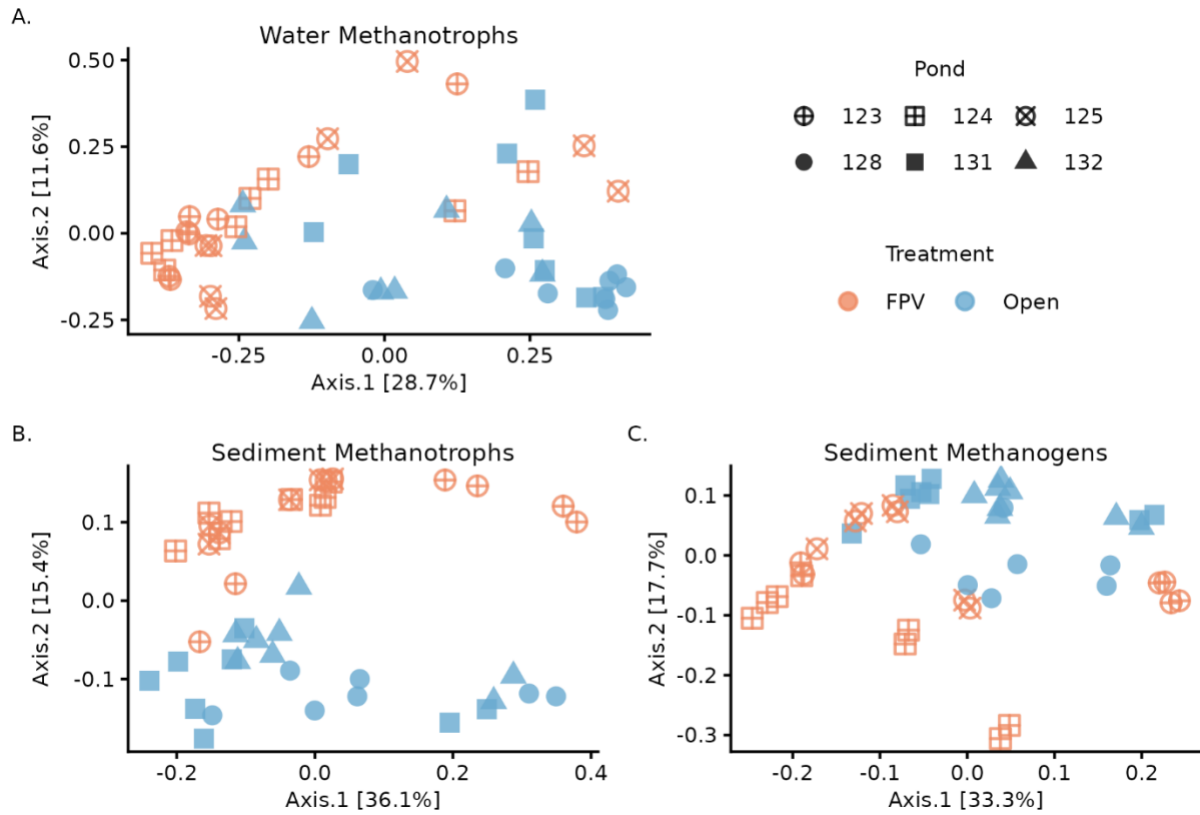


Fig. S5: Bray-Curtis PCoA of (A) water methanotrophs, sediment (B) methanotrophs and (C) methanogen communities within ponds with FPV (“FPV”) and ponds without FPV (“Open”); Sediment replicates and are shown separately.

Water Methanogen ASVs Enriched in Open Ponds

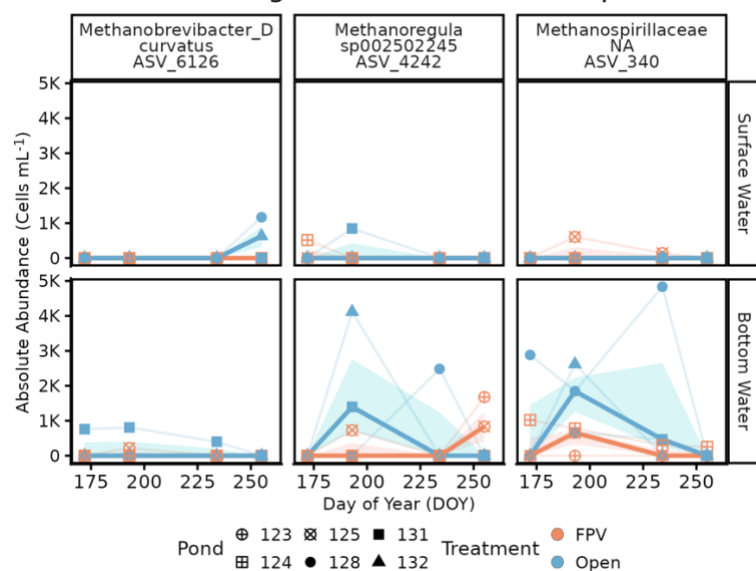


Fig. S6. Time-averaged differential abundance of water-column methanogen ASVs under non-FPV pond conditions. Absolute abundances (cells mL⁻¹) of water-column methanogen ASVs identified as enriched in open ponds using time-averaged ANCOM-BC2 analyses are shown across the sampling season.

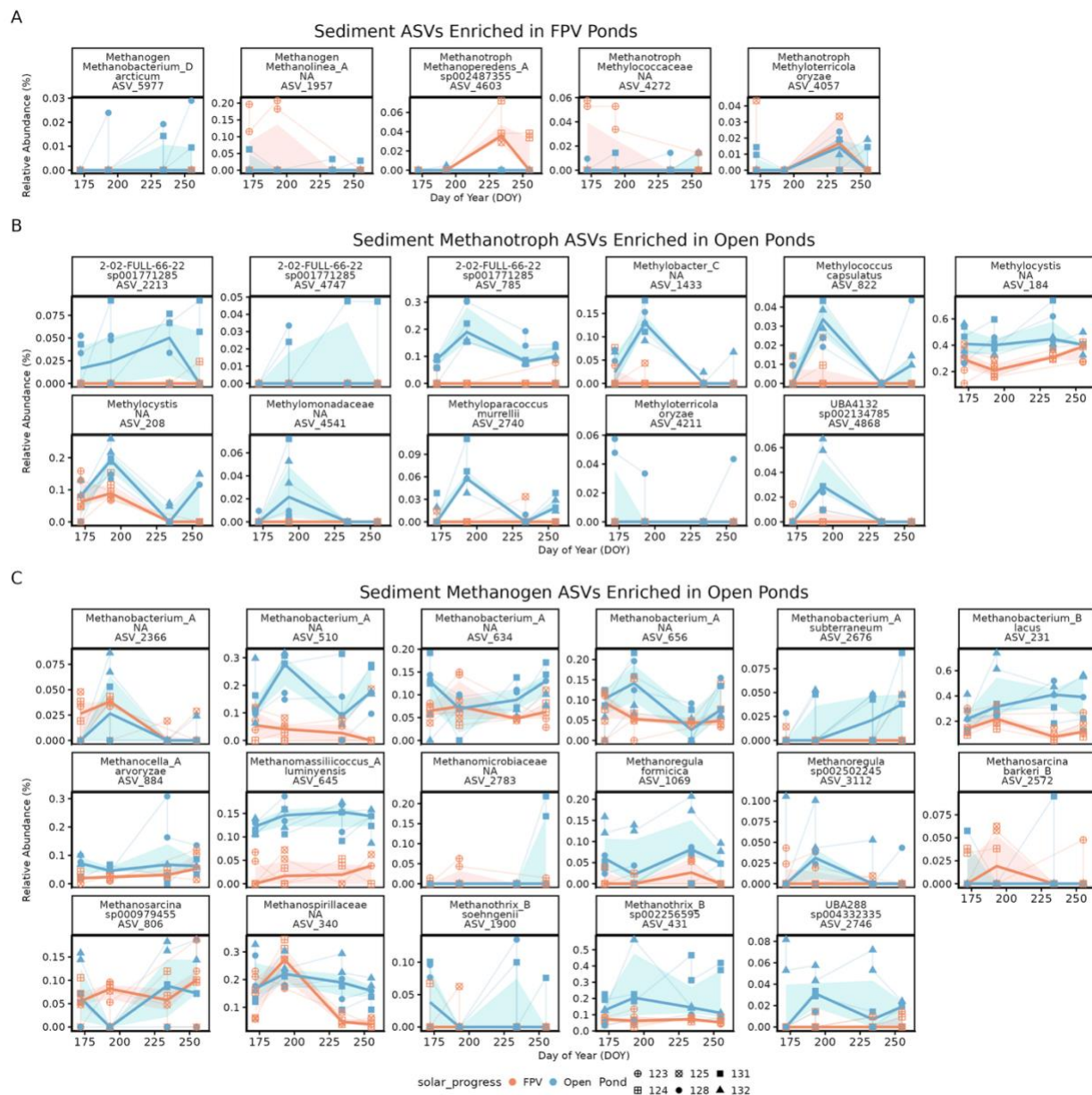


Fig. S7. Time-averaged differential abundance of sediment methane-cycling ASVs under FPV and non-FPV pond (“Open”) conditions. Sediment methane-cycling ASVs identified as differentially abundant using time-averaged ANCOM-BC2 analyses. (A) ASVs enriched in FPV ponds, (B) methanotroph ASVs enriched in open ponds, and (C) methanogen ASVs enriched in open ponds. Relative abundance (%) of each ASV is shown across day of year (DOY). ASVs lacking genus-level taxonomic resolution are labeled at the family level in facet titles (ASV_340, ASV_2783, ASV_4541, ASV_2366, ASV_634).

Seasonal FPV Effects on Sediment Methanotroph ASVs

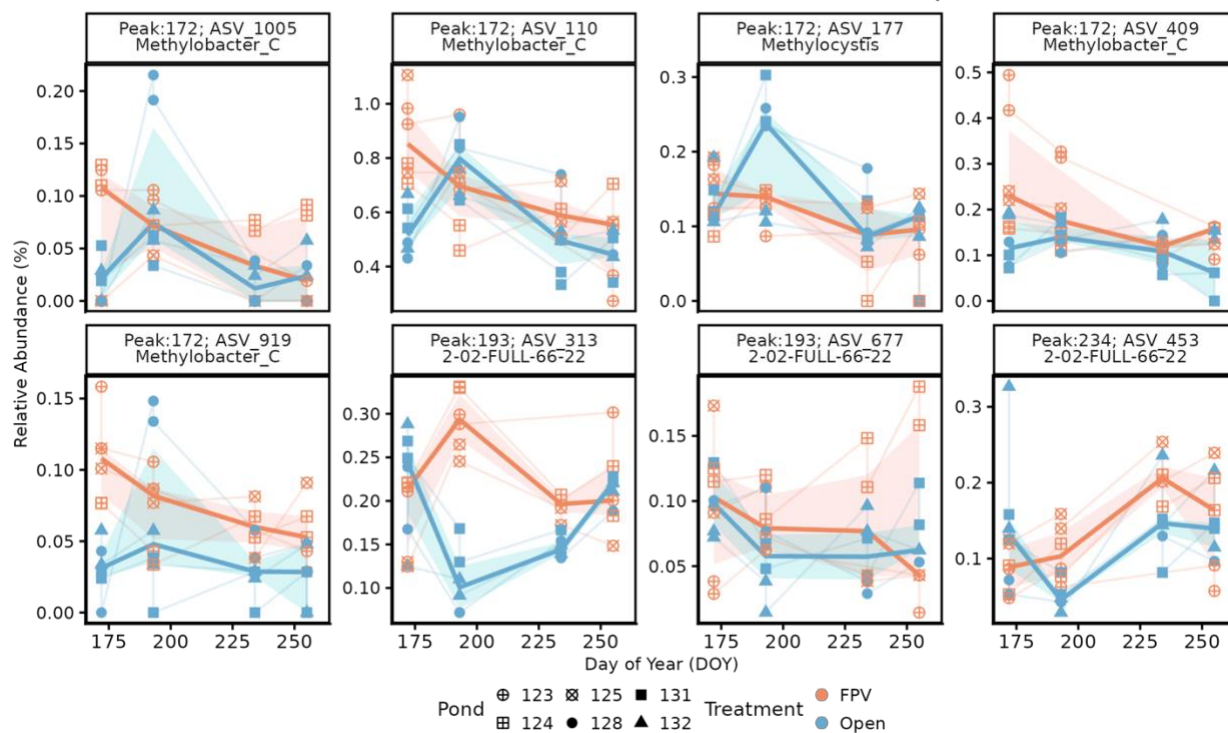


Fig. S8. Seasonally explicit enrichment of sediment methanotroph ASVs under FPV conditions. Relative abundance (%) of selected sediment methanotroph ASVs across the sampling season, highlighting ASVs with peak FPV-associated enrichment identified using a seasonally explicit (time-resolved) screening approach rather than time-averaged differential abundance testing. Taxa include Gammaproteobacteria (Type I; *Methylobacter_C*), Alphaproteobacteria (Type II; *Methylocystis*), and anaerobic nitrite-dependent methanotrophs (Type III; *Methyloirabilales*, 2-02-FULL-66-22). Community-level analyses did not detect a significant $\text{FPV} \times \text{time}$ effect for sediment methanotrophs.